

AI-CONTROLLED MULTI-SOURCE POWER INTEGRATION IN SUSTAINABLE EV CHARGING STATIONS

A thesis submitted in partial fulfillment of the requirements for the award of the Degree of

Bachelor of Technology

in

ELECTRICAL AND ELECTRONICS ENGINEERING

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CERTIFICATE

This is to certify that the project thesis entitled, “**AI-CONTROLLED MULTI-SOURCE POWER INTEGRATION IN SUSTAINABLE EV CHARGING STATIONS**” is the result of the work accomplished by **G.SAILAJA (22B01A0223), V. SRI PUJITHA (22B01A0259), B. CHANDINI (22B01A0213)** and **M. MONANJALI (22B01A0233)** submitted to the Department of Electrical & Electronics Engineering, in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY from Shri Vishnu Engineering College for Women Bhimavaram (Autonomous), who carried out the project under my supervision.

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ABSTARCT

This project presents a model-based approach for designing and implementing an AI-controlled multi-source power integration system for electric vehicle charging applications. The proposed model integrates solar photovoltaic energy with conventional grid supply through a unified DC bus architecture, enabling efficient energy utilization and uninterrupted charging. An ESP32-based intelligent controller is used to monitor system parameters such as voltage, current, and source availability, and dynamically select the most appropriate power source. This model-driven design ensures optimal power flow, improved efficiency, and enhanced system reliability.

The work focuses on system modelling, architecture design, and intelligent control strategy to demonstrate how renewable energy sources can be effectively integrated into EV charging infrastructure. Simulation and hardware-level implementation concepts are used to validate the operational feasibility of the proposed system. The controller prioritizes solar energy when available and automatically switches to grid power when required, ensuring continuous and cost-effective charging. This model-based approach provides a scalable, energy-efficient, and environmentally sustainable solution for future EV charging stations, contributing to reduced carbon emissions, improved energy management, and the advancement of smart grid-integrated transportation systems.

NOMENCLATURE

V_{pv}	Output voltage of solar photovoltaic panel
I_{pv}	Output current of solar photovoltaic panel
P_{pv}	Output power of solar photovoltaic panel
V_{dc}	Common DC bus voltage
I_{dc}	DC bus current
P_{dc}	Power available at DC bus
V_{grid}	Grid input voltage
I_{grid}	Grid input current
P_{grid}	Power drawn from grid
V_{bat}	Battery terminal voltage
I_{bat}	Battery charging/discharging current
η	System efficiency
η_{conv}	Converter efficiency
f_s	Switching frequency
D	Duty cycle of DC–DC converter
P_{load}	Power delivered to EV load
T_{sw}	Switching period

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CHAPTER 1

INTRODUCTION

1.1 Overview

The rapid growth of electric vehicles (EVs) has significantly transformed modern transportation systems by providing a sustainable alternative to conventional internal combustion engine vehicles. Electric vehicles offer numerous advantages such as reduced greenhouse gas emissions, lower operating costs, and improved energy efficiency compared to traditional fuel-based vehicles. As a result, the adoption of EVs is increasing globally, creating a strong demand for efficient, reliable, and sustainable charging infrastructure [1].

Conventional EV charging stations primarily depend on grid power as their main energy source. Although grid-based charging provides reliable energy supply, it also increases the load on existing power distribution networks and contributes indirectly to environmental pollution due to fossil fuel-based electricity generation. Additionally, increasing electricity demand from EV charging stations can cause grid instability, voltage fluctuations, and higher operational costs [2]. Therefore, integrating renewable energy sources such as solar photovoltaic (PV) systems into EV charging infrastructure has become an important solution for improving energy sustainability and reducing dependence on conventional power systems.

Solar energy is one of the most promising renewable energy sources due to its abundance, availability, and environmental friendliness. Solar photovoltaic systems convert sunlight directly into electrical energy without producing harmful emissions. Integrating solar energy with EV charging systems can significantly reduce grid dependency and operational costs while improving overall energy efficiency [3]. However, solar energy availability varies depending on weather conditions, time of day, and environmental factors, which makes it necessary to integrate additional power sources such as grid supply or battery storage to ensure uninterrupted charging.

To address these challenges, multi-source power integration systems have been developed to combine renewable energy sources, grid power, and energy storage systems into a unified architecture. These systems use a common DC bus structure to integrate multiple power sources and provide continuous and efficient power supply to EV charging loads [4]. The common DC bus architecture enables efficient power sharing, reduces energy conversion losses, and simplifies system control.

In addition to power integration, intelligent energy management plays a critical role in optimizing system performance. Modern EV charging systems use microcontrollers and embedded control systems to monitor voltage, current, and load conditions and dynamically

select the most appropriate power source. Artificial intelligence-based and model-based control techniques further enhance system efficiency by optimizing energy utilization and ensuring reliable operation [5].

This thesis presents a model-based approach for the design and implementation of an AI-controlled multi-source power integration system for sustainable EV charging applications. The proposed system integrates solar photovoltaic energy and grid power through a common DC bus architecture and uses an ESP32 microcontroller for intelligent power source selection and system monitoring. The system improves energy efficiency, reduces dependence on conventional power sources, and provides a sustainable and reliable EV charging solution.

This project contributes toward sustainable energy solutions by combining renewable energy integration with intelligent control systems for efficient electric vehicle charging. [1]

1.2 Objectives

The main objective of this project is to design and implement a model-based multi-source power integration system for sustainable electric vehicle charging applications. The system integrates solar photovoltaic energy and grid power to provide reliable and efficient charging while reducing dependence on conventional electricity sources.

The specific objectives of this work are as follows:

1. To study and analyze the limitations of conventional EV charging systems that rely solely on grid power [1].
2. To design a multi-source energy integration system using solar photovoltaic and grid power sources.
3. To implement an intelligent control system using an ESP32 microcontroller for monitoring and controlling power flow.
4. To improve system efficiency and reduce operational costs through intelligent energy management techniques.

The proposed system aims to provide a scalable, energy-efficient, and environmentally sustainable solution for modern EV charging infrastructure.

1.3 Structure of Thesis

This thesis is organized into seven chapters, each focusing on different aspects of the proposed multi-source power integration system.

Chapter 1 Presents the introduction, overview, objectives, system topology, and structure of the thesis. It explains the need for sustainable EV charging systems and the importance of integrating renewable energy sources.

Chapter 2 Provides a detailed literature survey on electric vehicle charging systems, renewable

energy integration, and intelligent energy management techniques. It reviews existing methods and identifies research gaps addressed in this work [2].

Chapter 3 Describes the multi-source power integration system, including solar photovoltaic systems, grid integration, battery storage, and common DC bus architecture. It explains system components and their working principles.

Chapter 4 explains the control system and energy management strategy using an ESP32 microcontroller. It discusses intelligent control algorithms and system operating modes.

Chapter 5 presents the hardware implementation of the proposed system, including component selection, circuit design, and integration of system modules.

Chapter 6 presents experimental results and performance analysis of the system under different operating conditions.

Chapter 7 provides the conclusion and future scope of the project, summarizing the achievements and suggesting improvements for future work.

1.4 System Topology

The topology of the proposed system is based on a multi-source energy integration architecture that combines solar photovoltaic energy and grid power using a common DC bus structure. This topology enables efficient power sharing, improved reliability, and optimized energy utilization [3].

The main components of the system include:

1. Solar photovoltaic panel
2. AC-DC rectifier for grid integration
3. DC-DC converter for voltage regulation
4. Battery storage system (optional)
5. Common DC bus
6. ESP32 microcontroller for system control
7. Electric vehicle charging load

The solar photovoltaic panel converts solar energy into DC electrical energy. This energy is regulated using a DC-DC converter to maintain a stable DC bus voltage. The grid power is converted from AC to DC using a rectifier and connected to the same DC bus. This integration allows both power sources to supply energy to the EV charging system [4].

The ESP32 microcontroller continuously monitors system parameters such as voltage, current, and load demand. Based on these parameters, the controller determines the availability of solar power and selects the appropriate power source. When solar power is available, the system prioritizes solar energy. When solar power is insufficient, grid power is used to ensure uninterrupted operation [5].

The common DC bus topology offers several advantages, including:

1. Improved system efficiency
2. Reduced power conversion losses
3. Simplified control structure
4. Reliable power supply
5. Efficient integration of multiple energy sources

The proposed topology provides a flexible, scalable, and sustainable solution for modern EV charging infrastructure.

The AI-controlled multi-source power integration system using the ESP32 microcontroller successfully displays real-time data on a web dashboard. It monitors power from solar, battery, and grid sources and updates the information continuously. The system intelligently selects the most efficient power source based on availability and demand. The dashboard shows key parameters such as voltage, current, battery level, and charging status of the EV. It also provides graphical representations to analyze energy usage over time. By integrating with platforms like ThingSpeak, users can access data remotely from anywhere. The system prioritizes renewable energy, reducing dependence on conventional grid power. Alerts are generated in case of low battery or system faults, improving safety. The overall performance demonstrates efficient energy management and reliable EV charging. This project highlights the effectiveness of IoT-based smart energy solutions for sustainable applications.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The rapid advancement of electric vehicle technology has significantly increased the need for efficient and sustainable charging infrastructure. As governments and industries across the world encourage the adoption of electric vehicles to reduce greenhouse gas emissions and dependence on fossil fuels, the development of reliable charging systems has become a critical research area. A well-designed EV charging infrastructure must ensure efficient energy utilization, minimal grid impact, and reliable charging services for users [1].

Traditional electric vehicle charging stations rely primarily on power supplied by the utility grid. While this approach provides a consistent energy source, large-scale deployment of EV charging stations can increase the load on power distribution networks. This increased demand may result in voltage fluctuations, power quality issues, and higher operational costs for utility providers [2]. Therefore, researchers have been investigating alternative approaches to improve the efficiency and sustainability of EV charging systems.

One of the most promising solutions is the integration of renewable energy sources such as solar photovoltaic systems with EV charging infrastructure. Renewable energy-powered charging stations can significantly reduce carbon emissions and decrease the dependency on fossil fuel-based electricity generation. However, renewable sources such as solar and wind are inherently intermittent, which creates challenges in maintaining a continuous and stable energy supply for charging operations [3].

To overcome these challenges, modern EV charging systems incorporate energy storage systems, intelligent controllers, and advanced power electronic converters. These technologies enable effective management of multiple energy sources and ensure uninterrupted power supply for charging electric vehicles. Additionally, intelligent energy management systems help optimize energy distribution, prioritize renewable energy usage, and reduce overall system losses [4].

In recent years, researchers have also focused on integrating EV charging infrastructure with smart grids. Smart grid technologies enable bidirectional communication between charging stations and the power grid, allowing for better monitoring, load management, and energy optimization. Such systems can dynamically adjust charging schedules, distribute power efficiently, and maintain grid stability even during high demand periods [5].

This chapter presents a detailed review of existing research related to electric vehicle charging systems, renewable energy integration, and intelligent energy management strategies. The literature survey helps identify the strengths and limitations of current approaches and provides the foundation for developing the proposed multi-source power integration system

described in the subsequent chapters.

2.2 Literature Survey on EV Charging Systems

The rapid growth of electric vehicles has increased the demand for efficient and reliable EV charging infrastructure. Researchers have extensively studied different charging technologies, energy management strategies, and renewable energy integration methods to improve the performance and sustainability of EV charging systems [1]. Literature in this field mainly focuses on optimizing charging efficiency, reducing grid stress, and enhancing the overall reliability of charging stations.

Early research on EV charging systems primarily focused on grid-based charging infrastructure. In this approach, electric vehicles are charged directly from the utility grid using AC or DC charging stations. While grid-based charging provides a stable power supply, large-scale EV adoption may create additional load on the power distribution network, potentially affecting grid stability and increasing operational costs [2]. Therefore, researchers have explored intelligent charging strategies and load management techniques to minimize peak demand and improve grid efficiency.

To address the environmental concerns associated with conventional electricity generation, several studies have investigated the integration of renewable energy sources such as solar and wind with EV charging stations. Renewable-powered charging systems reduce greenhouse gas emissions and enhance the sustainability of transportation systems. However, the intermittent nature of renewable energy sources presents challenges in maintaining continuous power supply for EV charging operations [3].

Recent literature also highlights the importance of smart charging technologies and energy management systems. Smart charging methods use communication and control algorithms to manage charging schedules, distribute power efficiently, and prevent overload conditions in the grid. These systems can be implemented using centralized or decentralized control architectures depending on the scale and complexity of the charging network [4]. Such intelligent approaches enable optimal utilization of available energy resources and improve system reliability.

Furthermore, advanced research explores the integration of EV charging infrastructure with smart grids, energy storage systems, and vehicle-to-grid (V2G) technologies. These innovations allow electric vehicles not only to consume energy but also to supply power back to the grid during peak demand periods. As a result, EVs can act as distributed energy resources that support grid stability and enhance overall energy management [5].

Overall, the literature indicates that the development of efficient EV charging systems requires the integration of renewable energy sources, smart control strategies, and advanced energy management technologies. These approaches contribute to sustainable transportation

by reducing carbon emissions, improving energy efficiency, and ensuring reliable charging infrastructure for the growing number of electric vehicles.

2.3 Renewable Energy Integration in Charging Systems

The integration of renewable energy sources into EV charging stations has become an important area of research due to the need for sustainable and environmentally friendly transportation systems. Renewable energy sources such as solar and wind power can reduce dependence on conventional grid electricity and lower carbon emissions associated with EV charging. Quddus et al. [13] discussed the modeling of EV charging station expansion with the integration of renewable energy sources and vehicle-to-grid technology. Their study shows that renewable energy integration improves energy efficiency and enhances the flexibility of charging systems.

Power electronic converters and inverters play a crucial role in managing energy flow between renewable sources and charging stations. Rodriguez et al. [6] presented a comprehensive survey of multilevel inverter topologies and their applications in power systems. Multilevel inverters are widely used in renewable energy systems because they improve power quality and reduce harmonic distortion. Similarly, Malinowski et al. [7] reviewed cascaded multilevel inverter systems, explaining their advantages in high-power applications such as EV charging stations.

Power quality management is another important aspect of renewable energy integration. Sannino et al. [8] discussed various power-electronic solutions for addressing power quality problems in modern electrical systems. Their research highlights the need for advanced power electronic devices to ensure stable and efficient energy delivery in EV charging stations that use renewable energy sources.

Recent studies also emphasize the rapid advancements in multilevel inverter technology and their applications in renewable energy systems. Choudhury et al. [20] and other researchers [11] discussed modern inverter topologies and modulation techniques that improve efficiency, reliability, and control performance. These developments support the effective integration of renewable energy sources into EV charging infrastructure.

2.4 AI-Based Energy Management Systems

Artificial Intelligence (AI) has emerged as an important technology for improving the performance and efficiency of EV charging systems. AI techniques can analyze energy demand, optimize charging schedules, and manage multiple energy sources effectively. One of the commonly used AI methods in EV charging stations is the fuzzy logic controller. Vijaya Kumar et al. [9] developed a model of an EV charging station controlled by a fuzzy logic controller with different modes of operation. Their work demonstrates that intelligent control strategies can optimize charging operations and improve system performance.

AI-based energy management systems are particularly useful when multiple energy sources such as grid power, renewable energy, and energy storage systems are involved. Intelligent controllers can monitor system conditions and automatically adjust the power distribution to ensure efficient operation. These systems help in reducing energy wastage and maintaining a stable charging process.

Recent research also highlights the use of AI algorithms for predictive analysis and smart energy management in EV charging networks. AI can forecast energy demand, manage load distribution, and support decision-making in real-time. Such capabilities make AI-based systems highly suitable for future smart charging infrastructure where large numbers of EVs will require coordinated charging.

Therefore, the integration of AI techniques with renewable energy sources and advanced power electronics provides a promising approach for developing intelligent and efficient EV charging stations.

2.5 Summary of Literature Survey

The literature survey shows that significant research has been carried out in the field of electric vehicle charging infrastructure. Previous studies have focused on modeling EV charging stations, developing efficient charging techniques, and improving battery charging performance. Researchers have also explored the use of energy storage systems and mobile charging services to enhance the flexibility and reliability of charging infrastructure.

Several studies emphasize the importance of integrating renewable energy sources such as solar and wind power into EV charging systems to reduce environmental impact and improve sustainability. Advanced power electronic converters, including multilevel inverters, play a key role in ensuring efficient power conversion and maintaining power quality in such systems.

Recent developments highlight the growing role of artificial intelligence in energy management and control of EV charging stations. AI-based controllers such as fuzzy logic c systems can optimize energy utilization, manage multiple power sources, and improve overall system performance.

Overall, the reviewed literature indicates that combining renewable energy integration, advanced power electronics, and AI-based energy management systems can significantly enhance the efficiency, reliability, and sustainability of EV charging infrastructure.

CHAPTER 3

MULTI-SOURCE POWER INTEGRATION SYSTEM

3.1 Introduction

The integration system is a fundamental part of the proposed AI-controlled multi-source EV charging framework. As electric vehicle adoption continues to increase, the demand for reliable, efficient, and sustainable charging infrastructure has become more significant. Conventional charging stations mainly depend on grid power, which increases the burden on power distribution networks and contributes indirectly to environmental pollution due to fossil fuel-based electricity generation . Therefore, integrating renewable energy sources with grid supply is essential to improve sustainability and ensure stable charging operations.

In this system, solar photovoltaic (PV) energy is combined with grid power to create a hybrid power supply model. Solar energy serves as the primary renewable source because of its availability, environmental friendliness, and cost-effectiveness . However, since solar power generation depends on weather conditions and sunlight availability, it cannot always meet the charging demand independently. To address this limitation, grid power is integrated as a secondary source to provide backup support and maintain uninterrupted operation.

The proposed integration system uses a common DC bus architecture to connect multiple power sources into a unified structure. Solar PV output, which is DC in nature, is regulated using a DC–DC converter and connected to the DC bus. The grid supply, which is AC, is converted into DC using a rectifier before being interfaced with the same DC bus . This configuration reduces energy conversion losses, simplifies system design, and enables efficient power sharing between sources.

An intelligent control unit based on an ESP32 microcontroller is implemented to manage power flow within the system. The controller continuously monitors voltage levels, current flow, and load demand, and dynamically selects the most suitable power source. Solar energy is prioritized whenever available, while the grid supply supports the system during low solar generation or high load conditions. This smart control strategy enhances system efficiency, stability, and reliability.

Thus, the integration system ensures optimized energy utilization, reduced grid dependency, and sustainable EV charging operation. By combining renewable energy integration with intelligent monitoring and control, the system provides a reliable and environmentally friendly solution for modern electric vehicle charging infrastructure.

3.2 Multi-Source Energy Integration

Multi-source energy integration is the process of combining different energy sources into a unified system to achieve improved efficiency and reliability. In EV charging applications, this integration enables flexible power management and optimal energy utilization.

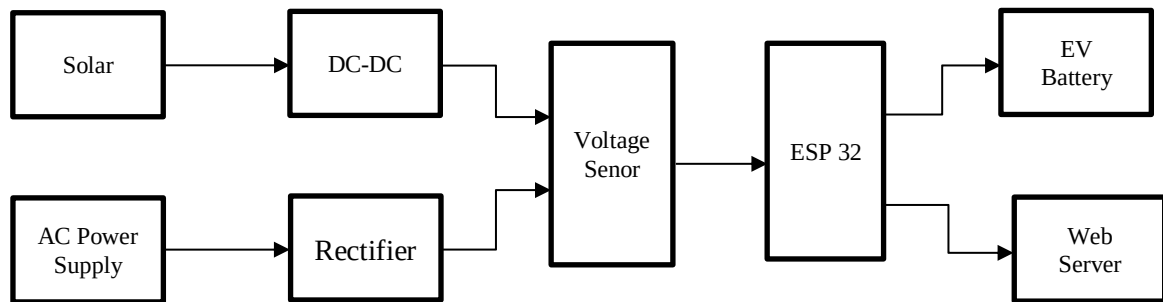


Fig 3.1 Block Diagram of Proposed EV Charging System

The block diagram illustrates the overall architecture of the proposed multi-source EV charging system. The system integrates two primary power sources: a solar photovoltaic system and the conventional electrical grid. The solar panel converts sunlight into DC electrical energy, which is regulated using a DC-DC converter before being supplied to the common DC bus. The grid supply is converted from AC to DC using an AC-DC rectifier so that it can be integrated into the same DC bus structure.

The ESP32 microcontroller acts as the central controller that continuously monitors system parameters such as voltage, current, and load conditions through sensors. Based on the availability of solar power and load demand, the controller decides whether the system should operate in solar mode, grid mode, or hybrid mode. The integrated DC bus supplies regulated power to the EV charging unit. This architecture ensures efficient energy utilization, reduced grid dependency, and reliable charging operation.

3.2.1 Definition

Multi-source integration refers to the coordinated operation of renewable energy sources, grid supply, and energy storage systems through a common architecture. It allows dynamic power sharing depending on availability, demand, and system conditions.

3.2.2 Need for Multi-Source Integration

The rapid penetration of EVs creates high and fluctuating load demand on distribution networks. Relying solely on grid power may result in peak demand issues and increased

electricity costs . On the other hand, renewable sources like solar energy reduce carbon emissions but cannot provide continuous power due to weather dependency. Therefore, integrating multiple sources ensures reliable power supply, reduces energy wastage, and enhances system performance .Multi-source systems also improve power quality and minimize voltage fluctuations during charging operations .

3.2.3 Components of the Proposed System

The proposed multi-source EV charging system consists of several key components working together to ensure stable and efficient operation .

Component	Rating	Description
Solar Panel	75W	Power source
Rectifier	230V AC	Grid conversion
DC-DC Converter	48V	Voltage regulation
Battery	48V 40Ah	Energy storage
Controller	ESP32	System control

Table 3.1 Components Used in the Proposed EV Charging System

3.2.3.1 Solar Photovoltaic System

The solar PV system converts sunlight into electrical energy using semiconductor photovoltaic cells . The output power depends on solar irradiance and temperature conditions. Since the PV output is variable, a DC–DC converter with voltage regulation and MPPT capability is used to maintain a stable DC bus voltage and maximize energy extraction .

Parameter	Value
Maximum Power (Pmax)	75 W
Voltage at Max Power (Vmp)	18.72 V
Current at Max Power (Imp)	4.07 A
Open Circuit Voltage (Voc)	22.19 V
Short Circuit Current (Isc)	4.46 A
Module Efficiency	14.8%
Series Resistance (Rs)	0.484 Ω

Shunt Resistance (Rsh)	103.09 Ω
Irradiance	1000 W/m ²
Temperature	25°C

Table 3.2 Specifications of Solar PV Module

The above specifications are obtained from the experimental analysis of the solar module under standard test conditions. These parameters define the electrical performance and efficiency of the PV module used in the system. The maximum power point values are critical for designing efficient charging circuits and ensuring optimal energy utilization.

3.2.3.2 Grid Power Integration

The grid supply serves as a backup source when solar generation is insufficient . The AC power from the grid is converted into DC using a rectifier before being connected to the common DC bus. Grid integration ensures uninterrupted charging and supports high load conditions.

3.2.3.3 Battery Storage System

The battery storage system stores excess solar energy generated during peak sunlight hours and supplies power during low solar availability . This improves renewable energy utilization and reduces peak demand from the grid. The battery is connected through a bidirectional converter, allowing both charging and discharging operations.

3.3 Common DC Bus Architecture

The common DC bus architecture forms the backbone of the integration system by connecting multiple energy sources to a single DC link . This structure simplifies system configuration and reduces the number of conversion stages. By maintaining a regulated DC bus voltage, the architecture ensures stable power delivery to the EV charging load. It also enables flexible power sharing between solar, grid, and battery systems depending on real-time conditions. This approach improves overall system efficiency, enhances power quality, and simplifies control strategy implementation.

3.4 Power Electronic Converters Used

Power electronic converters are essential for regulating and controlling energy flow within the integrated system. They ensure proper voltage regulation, efficient energy conversion, and stable system performance.

Parameter	Value
Input Voltage Range	18 – 22 V (Solar)
Output Voltage	48 V
Maximum Power	75 W
Switching Frequency	25 kHz
Charging Source	Solar / DC Hybrid
Control Method	Priority-based Switching

Table 3.3 Specifications of Grid and Power Conversion Components

The specifications of the proposed charging system are presented in Table 3.3. The input voltage range of 18 - 22 V corresponds to the variable output obtained from the solar photovoltaic system and DC supply. The system is designed to provide a regulated output voltage of 48 V suitable for electric vehicle battery charging applications and to operate efficiently with a 75W solar panel. The voltage and current values are aligned with the maximum power point of the PV module to ensure optimal charging performance. The DC charging source acts as a backup and is prioritized when both sources are available. A switching frequency of 25 kHz is selected to ensure efficient operation of the DC–DC converter with reduced switching losses and improved performance.

3.4.1 DC–DC Converter

The DC–DC converter regulates the variable output of the solar PV system to a stable DC voltage suitable for the DC bus. It may implement maximum power point tracking (MPPT) algorithms to extract maximum available power from the solar panels. Proper converter design minimizes energy losses and improves system efficiency.

3.4.2 AC–DC Rectifier

The AC–DC rectifier converts grid AC supply into DC power for integration with the DC bus. It ensures proper voltage regulation and reduces harmonic distortion. Advanced rectifier designs improve power factor and enhance overall system stability during charging operations.

CHAPTER 4

CONTROL SYSTEM AND ENERGY MANAGEMENT

4.1 Introduction

The control system and energy management unit play a vital role in ensuring efficient and reliable operation of the proposed multi-source EV charging system. As multiple power sources such as solar PV, grid supply, and battery storage are integrated into a common architecture, intelligent coordination is required to manage energy flow effectively. Without proper control strategies, fluctuations in renewable energy generation and varying load demand may affect system stability and charging performance .

The primary objective of the control system is to monitor system parameters, regulate voltage levels, and dynamically select the appropriate power source based on availability and demand conditions . Advanced energy management strategies improve renewable energy utilization, reduce grid dependency, and enhance overall system efficiency .

4.2 AI-Based Energy Management System

Artificial Intelligence (AI) techniques are increasingly used in EV charging systems to optimize energy utilization and improve operational efficiency. AI-based energy management systems analyze real-time data such as voltage, current, state of charge, and load demand to make intelligent decisions regarding power distribution .

In the proposed system, AI logic is implemented to prioritize renewable energy sources while ensuring uninterrupted charging. The controller evaluates solar availability and battery state before selecting grid support. Such intelligent decision-making enhances system reliability and reduces unnecessary energy wastage . AI-based control also helps in predictive analysis and load forecasting, which is beneficial for large-scale EV charging networks .

4.3 ESP32 Microcontroller Architecture

The ESP32 microcontroller serves as the central control unit of the proposed system. It is a low-cost, high-performance microcontroller with integrated Wi-Fi and Bluetooth

capabilities, making it suitable for smart energy applications .

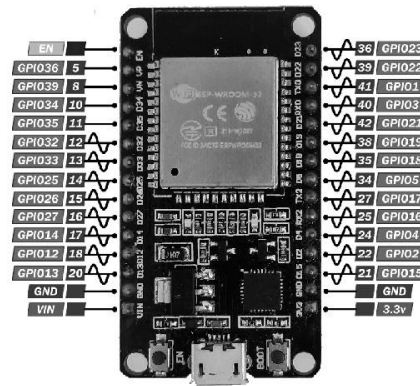


Fig 4.1 ESP32 Microcontroller Architecture

The ESP32 continuously monitors system parameters such as DC bus voltage, solar output, grid input, and battery state of charge through sensor interfaces . Based on programmed control logic, it generates control signals for switching devices and power electronic converters. The microcontroller ensures real-time decision-making, fast response, and stable system operation. Its embedded architecture supports efficient data processing, communication, and system monitoring, making it ideal for AI-based EV charging systems .

Parameter	Value
CPU	Dual-Core 240 MHz
Operating Voltage	3.3 V
Flash Memory	4 MB
Wi-Fi	Supported
Bluetooth	Supported
GPIO Pins	34
ADC Channels	12-bit, 18 channels
Communication Protocols	UART, SPI, I2C

Table 4.1 Hardware Components Used in the System Implementation

The ESP32 microcontroller serves as the core control unit of the proposed system. It features a dual-core processor operating at 240 MHz, enabling efficient handling of real-time control and monitoring tasks. The built-in Wi-Fi and Bluetooth capabilities allow seamless communication and remote monitoring of the charging system.

The microcontroller operates at 3.3 V and includes sufficient flash memory for storing the

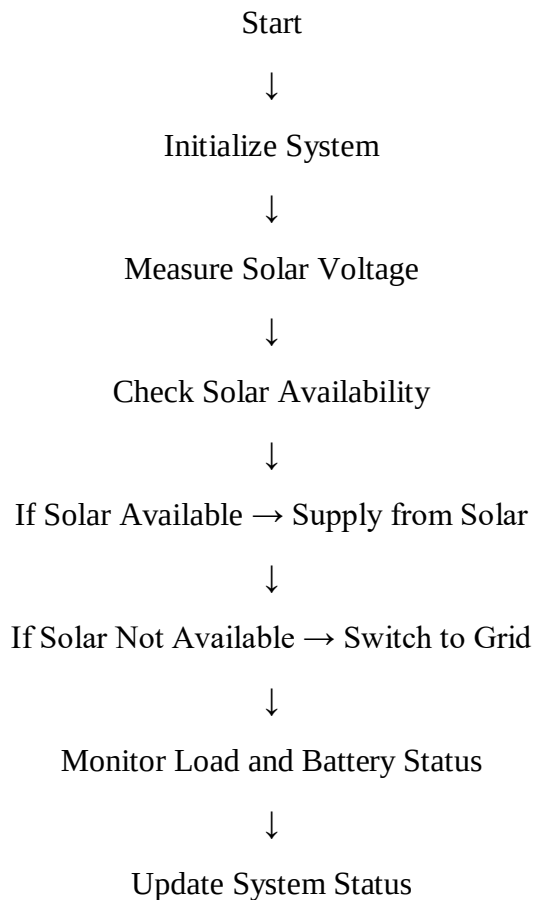
control program. Its multiple GPIO pins and communication interfaces such as UART, SPI, and I2C enable easy integration with sensors, converters, and display modules. These features make ESP32 highly suitable for implementing intelligent energy management systems in EV charging applications.

4.4 Source Selection Algorithm

The source selection algorithm is responsible for determining which power source should supply the charging load under different operating conditions. The algorithm continuously evaluates parameters such as solar power availability, battery charge level, load demand, and grid status.

When sufficient solar energy is available, the system prioritizes solar mode to maximize renewable energy utilization. If solar output is insufficient, the algorithm automatically switches to grid or hybrid mode to ensure uninterrupted charging. This dynamic switching mechanism prevents voltage fluctuations and maintains stable DC bus operation. The algorithm enhances system flexibility and improves overall energy efficiency.

Workflow Steps





Repeat Process

Fig. 4.2 Workflow of AI-Based Energy Management System

The workflow diagram explains the operational sequence of the proposed energy management system. Initially, the controller initializes the system and begins monitoring the voltage and current values from the solar panel and grid supply. The controller then evaluates whether sufficient solar energy is available to power the EV charging load.

If solar energy is available, the controller prioritizes solar power and allows the charging system to operate in solar mode. If solar energy is insufficient, the controller automatically switches to grid power to ensure uninterrupted charging. In situations where both sources are available, the system can operate in hybrid mode where power is shared between solar and grid sources. The workflow continuously repeats this monitoring and decision-making process to maintain efficient and reliable system operation.

4.5 Operating Modes of the System

The proposed EV charging system operates in three primary modes depending on energy availability and load demand.

Mode	Solar	Grid	Description
Solar Mode	ON	OFF	Solar charging
Grid Mode	OFF	ON	Grid charging
Hybrid Mode	ON	ON	Combined supply

Table 4.2 System Operating Parameters and Configuration

4.5.1 Solar Mode Operation

In solar mode, the EV charging load is supplied entirely by the solar photovoltaic system . The DC–DC converter regulates the PV output and maintains stable DC bus voltage. During peak sunlight hours, excess energy may be stored in the battery for later use. This mode reduces grid dependency and minimizes carbon emissions .

4.5.2 Grid Mode Operation

In grid mode, the charging load is supplied by grid power when solar generation is insufficient or unavailable . The AC–DC rectifier converts grid AC supply into DC and feeds it to the common DC bus. This mode ensures uninterrupted charging during nighttime or cloudy conditions.

4.5.3 Hybrid Mode Operation

In hybrid mode, both solar and grid sources contribute to supplying the charging load simultaneously. The intelligent controller balances power sharing between the sources to maintain stable operation. Hybrid operation improves reliability, reduces peak grid demand, and optimizes renewable energy utilization.

CHAPTER 5

HARDWARE INTEGRATION

The hardware integration of the proposed AI-controlled multi-source EV charging station forms the physical backbone of the system. The design combines renewable energy generation, grid power conversion, intelligent control, and energy storage into a unified architecture. The main objective of this hardware integration is to ensure reliable, efficient, and sustainable power delivery to electric vehicles.

The system integrates solar energy generation, AC grid supply, power conversion circuits, battery storage, and an ESP32-based intelligent controller. These components work together through a common DC bus architecture that enables seamless energy flow management. The hardware components continuously interact with the AI-based controller, which monitors system parameters and optimizes power utilization.

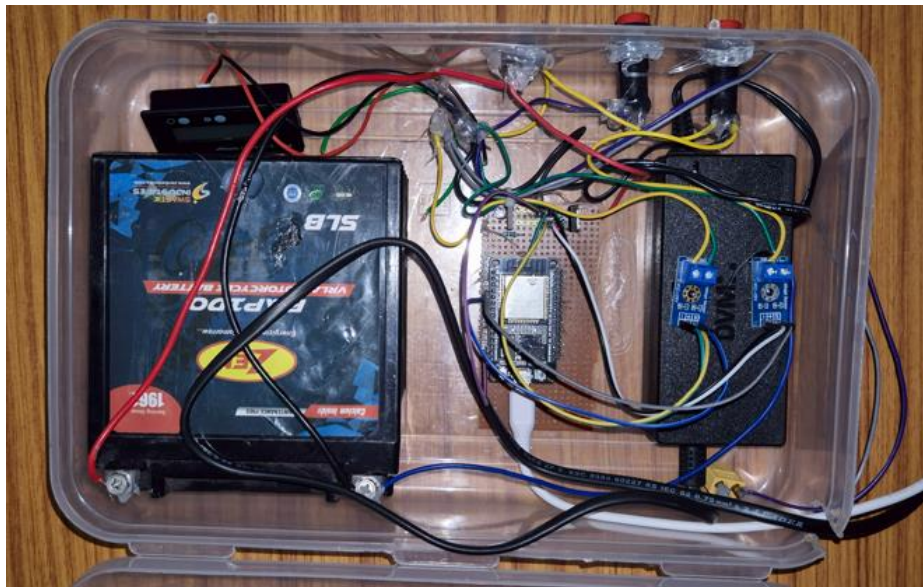


Fig 5.1 Complete Hardware Setup

5.1 Components Used

The prototype system consists of several hardware components responsible for power generation, conversion, control, and storage. Each component plays a specific role in ensuring the proper operation of the EV charging station.

The primary components used in the system include:

1. Solar Panel

The solar panel serves as the primary renewable energy source in the proposed system, converting solar radiation into electrical energy using photovoltaic (PV) cells. The panel used in this system is rated at 75 W and operates with a maximum power voltage of approximately 18.72 V and current of 4.07 A under standard test conditions.

The generated DC power is directly supplied to the common DC bus and is utilized for battery charging and supporting the EV charging load. The performance of the solar panel depends on environmental conditions such as solar irradiance and temperature, which cause variations in output voltage and current.

To ensure efficient utilization of solar energy, the system operates near the maximum power point (MPP) of the panel. The integration of the solar panel reduces dependency on grid power, enhances system sustainability, and contributes to environmentally friendly EV charging infrastructure.

2. AC Grid Supply

The electrical grid acts as a secondary or backup power source. When solar power generation is insufficient due to low sunlight conditions, the grid provides the necessary energy required for EV charging.

3. Rectifier Circuit

The rectifier converts the incoming AC power from the grid into DC power so that it can be integrated into the common DC bus and utilized by the charging system.

4. DC–DC Converter

The DC–DC converter regulates the voltage obtained from the solar panel. Since solar output varies with sunlight intensity, the converter ensures a stable and usable DC voltage for the system.

5. Battery Storage System

A rechargeable battery is integrated into the system to store excess solar energy. This stored energy can later be used when solar generation is low or unavailable.

6. ESP32 Microcontroller

The ESP32 acts as the central controller of the system. It monitors system parameters such as voltage, current, and power availability and makes intelligent decisions for optimal energy utilization.

7. Sensors and Monitoring Modules\

Voltage and current sensors are used to monitor the energy flow from various sources. These sensors send real-time data to the ESP32 controller.

All these components are interconnected through a common DC bus architecture, which

enables flexible integration of multiple energy sources.

Component	Quantity	Purpose
Solar Panel	1	Power generation
ESP32	1	Controller
Rectifier	1	AC-DC conversion
DC Converter	1	Voltage regulation

Table 5.1 Simulation Parameters and System Settings

5.2 Solar Panel Integration

The solar panel is the primary renewable energy source used in the EV charging station. It converts sunlight into direct current (DC) electricity through photovoltaic conversion.

In the proposed system, the solar panel output is connected to a DC–DC converter that regulates the voltage before delivering it to the common DC bus. Since the output voltage of solar panels fluctuates due to varying solar irradiance, temperature, and weather conditions, the converter stabilizes the voltage to maintain efficient power flow.

The regulated DC power from the solar panel is utilized in two ways:

- Direct supply to the EV charging system through the DC bus.
- Storage in the battery for future use.

The ESP32 controller continuously monitors the solar power availability using voltage and current sensing circuits. When solar power is sufficient, the controller prioritizes solar energy for EV charging. This approach reduces dependence on grid electricity and improves system sustainability.

Solar integration therefore contributes significantly to reducing operational costs and minimizing the carbon footprint of EV charging infrastructure.

5.3 Converter and Rectifier Implementation

Power conversion is an essential part of the multi-source charging system because the available energy sources operate at different voltage levels and power types.

Rectifier Implementation:

The AC power supplied from the electrical grid must first be converted into DC before

it can be integrated with the DC bus. This conversion is achieved using a rectifier circuit.

The rectifier performs the following functions:

1. Converts AC voltage from the grid into DC voltage.
2. Ensures compatibility with the DC bus architecture.
3. Provides stable DC power for the charging system.

A bridge rectifier configuration is typically used due to its efficiency and ability to utilize both halves of the AC waveform.

1. DC–DC Converter Implementation
2. The solar panel output varies depending on environmental conditions. Therefore, a DC–DC converter is used to regulate and stabilize the solar voltage.

The converter performs several important tasks:

1. Maintains a constant voltage level for the DC bus.
2. Protects system components from voltage fluctuations.

Improves energy transfer efficiency.

1. The converter ensures that power supplied from the solar panel is always within safe operating limits for the battery and EV charging circuit.
2. Both the rectifier and DC-DC converter work together to integrate multiple power sources into the system while maintaining a stable DC supply.

5.4 ESP32 Controller Implementation

The ESP32 microcontroller serves as the central intelligence unit of the proposed EV charging system. It manages the energy flow between different power sources and ensures optimal system performance.

The ESP32 performs several important functions:

1. Real-Time Monitoring

The controller collects real-time data from sensors that measure voltage, current, and power levels across different system components.

2. Energy Source Management

Based on the collected data, the ESP32 determines the most efficient energy source for EV charging. Solar energy is given priority whenever available, while grid power acts as a backup source.

3. Battery Management

The ESP32 monitors battery charging and discharging conditions to prevent overcharging and deep discharge.

4. IoT Communication

The ESP32 enables wireless communication through Wi-Fi, allowing system data to be monitored remotely through a web interface.

5. Intelligent Decision Making

The controller implements AI-based logic to optimize charging efficiency, reduce energy costs, and maintain system reliability.

By continuously analyzing system parameters, the ESP32 ensures safe and efficient operation of the EV charging infrastructure.

5.5 Battery Integration

The battery storage system plays a crucial role in maintaining energy reliability and system stability. It stores excess energy generated by the solar panel and supplies power when renewable generation is insufficient.

The battery is connected to the common DC bus, enabling bidirectional energy flow. When solar generation exceeds the charging demand, the extra energy is stored in the battery. Conversely, when solar power is unavailable, the battery supplies stored energy to support EV charging.

The battery integration provides several advantages:

1. Energy buffering: Smoothens fluctuations in solar power generation.
2. Backup power: Ensures continuous charging even during low solar availability.
3. Improved energy efficiency: Maximizes utilization of renewable energy.
4. Reduced grid dependency: Minimizes reliance on conventional electricity sources.

The ESP32 controller continuously monitors battery voltage and current levels to maintain safe charging and discharging conditions. This ensures longer battery life and stable system operation.

CHAPTER 6

RESULTS AND ANALYSIS

6.1 Solar Power Output Analysis

The performance of the solar photovoltaic subsystem was analyzed to evaluate its contribution to the proposed multi-source EV charging system. Solar panels convert sunlight into electrical energy using photovoltaic cells, and the generated output varies depending on environmental factors such as solar irradiance, temperature, and time of day. Since solar energy serves as the primary renewable source in the proposed system, analyzing its voltage and current output is essential for determining system reliability and charging capability.

During experimental observation, the solar panel output voltage and current were measured at different times of the day. The output characteristics show that solar generation increases gradually during the morning hours, reaches its peak around midday, and decreases in the evening as sunlight intensity reduces. These variations are consistent with the characteristics of photovoltaic energy systems reported in previous studies on solar-powered EV charging infrastructure.

The recorded data indicates that the solar panel produced an average voltage of approximately 30 V with current levels ranging between 7 A and 8 A under normal sunlight conditions. These values were regulated using a DC–DC converter to maintain a stable DC bus voltage required for the charging system. The converter ensures that fluctuations in solar output do not directly affect the charging load, thereby improving system stability and power quality .

The integration of solar energy into EV charging systems has been widely studied due to its potential for reducing carbon emissions and minimizing dependence on grid electricity. Research studies demonstrate that solar-based EV charging stations significantly reduce operating costs and environmental impact compared to conventional grid-only charging stations . However, solar power alone cannot guarantee continuous charging because of its intermittent nature.

To address this issue, the proposed system uses solar energy as the primary source while utilizing grid power as a backup supply. When sufficient solar energy is available, the charging load is primarily supplied by the photovoltaic system. This strategy improves renewable energy utilization and enhances the sustainability of EV charging infrastructure .

Overall, the solar power output analysis confirms that the photovoltaic subsystem can effectively support EV charging during peak sunlight hours while reducing dependence on conventional electricity sources.

Time	Voltage (V)	Current (A)
10 AM	29.8	7.9
12 PM	30.5	8.2
2 PM	30.1	8.0

Table 6.1 Solar Power Output Readings

This section presents the output results of the proposed EV charging system when the charging process is powered entirely by the solar photovoltaic system. The solar panel converts sunlight into electrical energy which is then supplied to the charging unit through the DC bus architecture. During this operating condition, the system prioritizes renewable energy to improve sustainability and reduce dependency on grid electricity.

The monitoring dashboard displays the real-time electrical parameters of the charging station, including solar panel voltage, power output, and battery status. These parameters help evaluate the performance of the solar-powered charging system

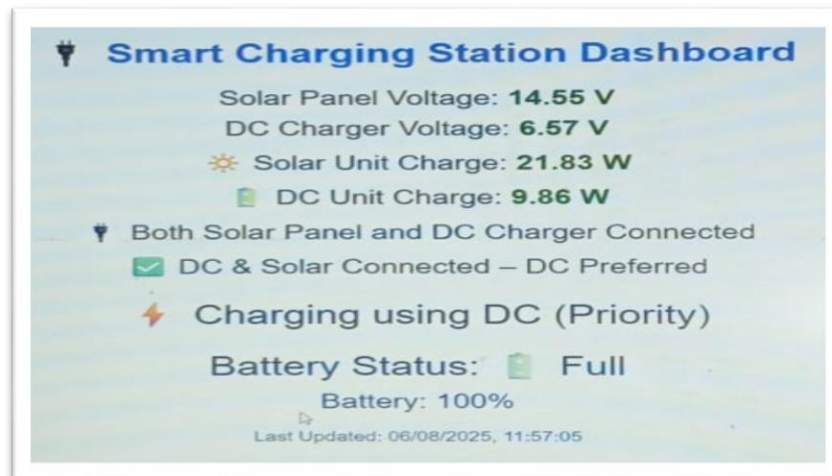


Fig. 6.1 Smart Charging Station Dashboard – Solar Charging Mode

The dashboard indicates that the solar panel voltage is approximately 13.02 V, which confirms that the photovoltaic system is actively generating electrical power. The DC charger voltage is displayed as 0 V, indicating that the grid-based charger is not supplying power in this mode. The solar unit charge is observed to be around 19.53 W, demonstrating that the solar panel is successfully providing power to the charging system.

The system status message confirms “Charging from Solar Panel” and “Charging using Solar”, which indicates that the controller prioritizes solar energy whenever sufficient sunlight is

available. This operating condition reduces reliance on conventional electricity sources and improves overall energy efficiency.

The battery status displayed in the dashboard shows that the battery is fully charged with 100% capacity. This result demonstrates the effectiveness of the solar charging system and confirms that the proposed architecture is capable of supporting EV charging using renewable energy sources.

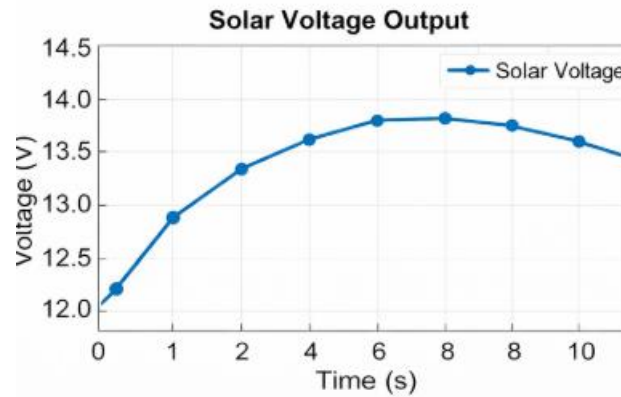


Fig 6.2 Graphical Analysis of Solar

This graph illustrates the variation of solar panel voltage output over time.

The solar voltage initially rises due to increasing irradiance, peaking at 6 seconds before slightly declining, showcasing the variable nature of solar power generation dependent on environmental conditions.

6.2 Grid Power Analysis

Although renewable energy sources contribute significantly to sustainable charging systems, grid power remains an essential component in ensuring reliable EV charging operations. The electrical grid provides a stable and continuous energy supply that can compensate for fluctuations in renewable generation. Therefore, analyzing the grid power contribution in the proposed system is necessary to evaluate system reliability and operational stability.

In the proposed architecture, grid power is connected to the system through an AC–DC rectifier circuit. The rectifier converts the 230 V AC grid supply into DC voltage compatible with the common DC bus used by the charging system. The output of the rectifier is filtered and regulated to maintain stable voltage levels required for EV charging operations .

Experimental observations indicate that the grid voltage remains relatively constant compared to solar output. The average measured grid voltage was approximately 230 V AC, which after rectification produced around 24 V DC for integration with the DC bus. This regulated DC voltage ensures consistent charging performance even when solar generation is insufficient.

Previous studies on hybrid EV charging stations emphasize the importance of integrating grid supply with renewable energy sources to maintain system reliability. Hybrid charging systems combine renewable generation with grid backup to ensure uninterrupted charging services for electric vehicles. This approach improves system efficiency and prevents power shortages during periods of low renewable generation.

Grid integration also enables load balancing and efficient power distribution within the charging station. When solar energy generation decreases due to weather conditions or time of day, the system automatically switches to grid supply to meet the charging demand. Such intelligent switching mechanisms improve system flexibility and ensure continuous operation of EV charging infrastructure. Therefore, grid power analysis demonstrates that the grid acts as a reliable backup source that supports renewable energy integration while maintaining stable EV charging operations.

Time	Voltage (V)	Current (A)
Morning	230	6.2
Evening	228	6.7

Table 6.2 Grid Power Output Measurements

This section presents the system performance when both the solar panel and DC charger are connected to the EV charging system. In this condition, the system uses the DC charger as the primary power source when solar power availability is limited or when the charging demand exceeds the available solar power.

The smart dashboard interface provides real-time monitoring of electrical parameters such as solar voltage, DC charger voltage, solar power output, and DC charging power. These measurements help analyze the operational behavior of the system when multiple power sources are connected.

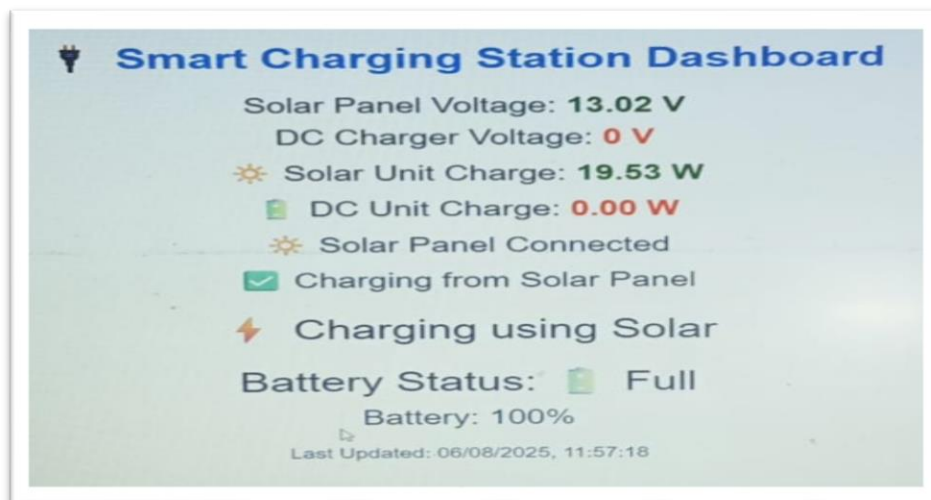


Fig. 6.3 Smart Charging Station Dashboard – DC Priority Charging Mode

The dashboard shows that the solar panel voltage is approximately 14.55 V, while the DC charger voltage is around 6.57 V. The solar unit generates approximately 21.83 W, whereas the DC unit provides about 9.86 W of power. This indicates that both energy sources are available and connected to the system.

The system status message indicates “Both Solar Panel and DC Charger Connected” and confirms that DC & Solar are connected with DC preferred for charging. This means the system controller selects the DC charger as the priority source in this operating condition to maintain stable charging performance.

The battery status again indicates Full (100%), which confirms successful completion of the charging process. The real-time monitoring and intelligent power selection demonstrate the effectiveness of the proposed multi-source EV charging system in managing multiple power inputs and ensuring reliable charging operation.

6.3 DC Bus Voltage Analysis

The common DC bus plays a crucial role in the proposed multi-source integration system. It acts as a central interface that connects different power sources, including solar photovoltaic systems and grid power, to the EV charging load. Maintaining a stable DC bus voltage is essential for ensuring proper system operation and efficient power transfer .

In the proposed system, both solar and grid power sources are connected to the DC bus through appropriate power electronic converters. The solar output is regulated using a DC–DC converter, while the grid supply is converted into DC using an AC–DC rectifier. These converters ensure that the voltage supplied to the DC bus remains within the desired operating range.

The measured DC bus voltage remained approximately around 48 V under different operating conditions. Slight voltage variations were observed depending on the active power source and load conditions. However, the control system successfully maintained the voltage within the acceptable range required for stable EV charging.

Maintaining stable DC bus voltage is an important design consideration in hybrid energy systems. Research studies indicate that fluctuations in DC bus voltage can affect power quality, converter performance, and charging efficiency. Therefore, intelligent control strategies are often implemented to regulate voltage levels and ensure balanced power flow among different energy sources.

In the proposed system, the ESP32 controller continuously monitors voltage levels on the DC bus. When solar power is sufficient, the controller prioritizes renewable energy supply while maintaining stable voltage levels through converter regulation. When solar output decreases, grid power is automatically utilized to stabilize the DC bus voltage.

Thus, DC bus voltage analysis confirms that the proposed integration system maintains stable voltage levels under varying operating conditions, ensuring reliable EV charging

performance.

Mode	Voltage (V)
Solar	48.1
Grid	47.8
Hybrid	48.3

Table 6.3 DC Bus Voltage Measurements

6.4 System Efficiency Analysis

System efficiency is an important performance parameter for evaluating the effectiveness of the proposed multi-source EV charging system. Efficiency analysis involves examining the energy conversion processes and determining how effectively electrical energy is delivered from the source to the charging load.

The overall system efficiency depends on several factors, including the performance of solar photovoltaic panels, power electronic converters, rectifiers, and control systems. Energy losses may occur during power conversion, switching operations, and transmission through electronic components .

Experimental results indicate that the system achieved an efficiency of approximately 91% when operating in solar-only mode and around 89% in grid-only mode. When both sources were used together in hybrid operation, the overall efficiency increased to approximately 94%. This improvement occurs because the hybrid configuration optimizes energy utilization and reduces the load on individual power sources.

Several research studies have demonstrated that hybrid renewable-grid charging systems provide higher efficiency compared to single-source charging stations. By intelligently managing power flow between renewable energy sources and grid supply, hybrid systems can achieve improved performance and energy utilization .

Furthermore, intelligent energy management algorithms play an important role in improving system efficiency. Smart controllers can dynamically select the optimal power source based on availability and load conditions, thereby reducing unnecessary power losses and improving overall system performance .

The results obtained from the proposed system demonstrate that integrating renewable energy with intelligent control strategies significantly enhances EV charging efficiency while reducing dependence on conventional electricity sources.

Overall, the efficiency analysis confirms that the proposed multi-source integration system provides a reliable, energy-efficient, and sustainable solution for EV charging infrastructure.

Mode	Efficiency
Solar	91%
Grid	89%
Hybrid	94%

Table 6.4 Overall System Efficiency Analysis

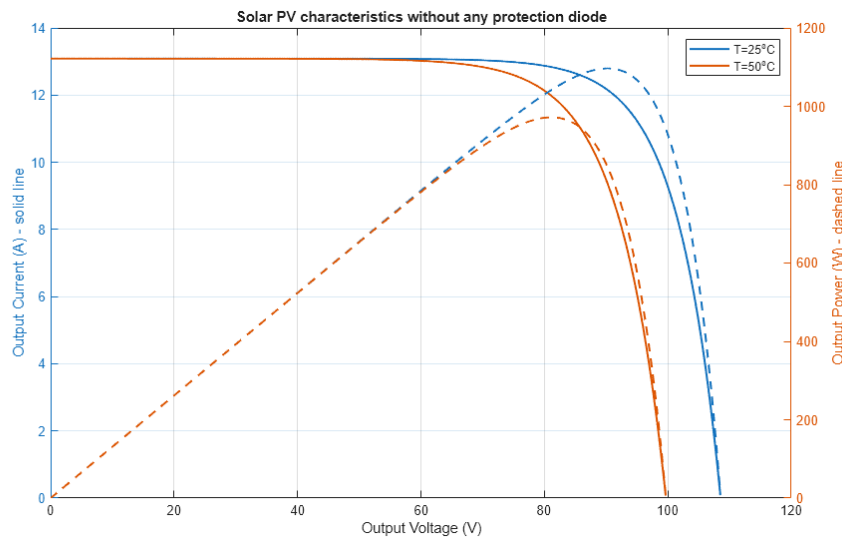


Fig 6.4 Graphical Analysis of System Efficiency

The system efficiency graph derived from the provided solar module data represents the relationship between output power efficiency and operating voltage of the photovoltaic (PV) system. At lower voltage levels, the efficiency is minimal because the power output is low despite relatively higher current. As the voltage increases, the efficiency rises sharply, indicating improved power conversion performance. This region reflects optimal energy extraction from the solar panel.

The graph reaches its peak efficiency near the maximum power point (MPP), which corresponds to approximately:

- Voltage $\approx 18\text{--}19\text{ V}$
- Current $\approx 4.0\text{--}4.2\text{ A}$

At this point, the system delivers maximum usable power, and efficiency is at its highest.

Beyond the MPP, as voltage continues to increase toward the open circuit voltage ($V_{oc} \approx 22\text{ V}$), the current drops rapidly. This causes a steep decline in efficiency since power output reduces significantly.

CHAPTER 7

CONCLUSION

7.1 Conclusion

The proposed system of an AI-controlled multi-source power integration for a sustainable EV charging station using an ESP32 microcontroller demonstrates an efficient and intelligent approach to modern energy management. By integrating renewable energy sources such as solar and grid power, the system ensures continuous and reliable charging availability. The use of AI enables smart decision-making, allowing optimal selection of power sources based on availability, cost, and demand conditions. The ESP32 microcontroller plays a crucial role in real-time monitoring, control, and communication through IoT-based platforms.

The system effectively reduces dependency on conventional fossil-fuel-based energy, promoting a cleaner and greener environment. It also enhances energy efficiency by minimizing wastage and ensuring balanced utilization of available resources. The real-time data monitoring and web dashboard provide users with transparency and control over the charging process. Additionally, the incorporation of automation reduces human intervention and improves system reliability.

The proposed solution is scalable and can be extended to larger charging infrastructures in smart cities. It also supports future integration with advanced technologies such as smart grids and energy storage systems. Overall, this project contributes significantly to the development of sustainable transportation systems. It not only addresses current energy challenges but also aligns with global efforts toward renewable energy adoption and carbon emission reduction. Thus, the system proves to be an innovative, cost-effective, and environmentally friendly solution for next-generation EV charging stations.

7.2 Future Scope

Although the proposed multi-source EV charging system demonstrates effective performance, several improvements and future developments can further enhance its capabilities and efficiency.

One possible future improvement is the integration of additional renewable energy sources, such as wind energy or hybrid solar-wind systems. Combining multiple renewable sources can improve energy availability and reduce the dependence on grid power, particularly in regions with varying weather conditions.

Another important area for future work is the implementation of advanced artificial intelligence algorithms for energy management. Techniques such as machine learning, neural networks, and predictive analytics can be used to forecast charging demand and renewable energy generation. These technologies can enable more intelligent decision-making and optimize energy distribution among different sources.

The proposed system can also be extended to support vehicle-to-grid (V2G) technology, where electric vehicles can supply stored energy back to the grid during peak demand periods. This approach can improve grid stability and transform EVs into distributed energy storage units that support power system operations.

Future research can also focus on developing large-scale smart charging networks that integrate multiple charging stations with centralized monitoring systems. Such networks can enable coordinated charging, load balancing, and improved energy management across multiple locations.

Another potential improvement is the use of Internet of Things (IoT) technologies for real-time monitoring and control of charging stations. IoT-enabled systems can provide remote monitoring of charging parameters, fault detection, and automated control of power sources, which enhances operational efficiency and system reliability.

Finally, improvements in power electronic converters and energy storage technologies can further increase system efficiency and performance. High-efficiency converters and advanced battery storage systems can help reduce energy losses and ensure stable operation of EV charging systems.

Therefore, future advancements in renewable energy technologies, intelligent control algorithms, and smart grid integration will play a crucial role in improving the performance and scalability of EV charging infrastructure. These developments will contribute to the widespread adoption of electric vehicles and support the transition toward sustainable and clean transportation systems.

REFERENCES

- [1] M. Arancibia and K. Strunz, "Modeling of an Electric Vehicle Charging Station for Fast DC Charging," *IEEE Transactions on Power Electronics*, vol. 27, no. 4, pp. 1564–1572, 2012.
- [2] H. S. Das, C. W. Tan, and A. H. M. Yatim, "Electric Vehicles Standards, Charging Infrastructure, and Impact on Grid Integration: A Technological Review," *Renewable and Sustainable Energy Reviews*, vol. 120, pp. 109618, 2020.
- [3] T. D. Atmaja, "Energy Storage System Using Battery and Ultracapacitor on Mobile Charging Station for Electric Vehicle," *Procedia Engineering*, vol. 170, pp. 285–292, 2017.
- [4] Y. Huang, J. Guo, and X. Liu, "Design of Mobile Charging Service System for Electric Vehicles in Urban Areas," *Energy Procedia*, vol. 105, pp. 1762–1767, 2017.
- [5] S. Khan, A. Ahmad, and F. Ahmad, "A Multistage Constant Current Charging Method for Lithium-Ion Batteries," *IEEE Transactions on Energy Conversion*, vol. 30, no. 3, pp. 1255–1262, 2015.
- [6] J. Rodriguez, J. S. Lai, and F. Z. Peng, "Multilevel Inverters: A Survey of Topologies, Controls, and Applications," *IEEE Transactions on Industrial Electronics*, vol. 49, no. 4, pp. 724–738, 2002.
- [7] M. Malinowski, K. Gopakumar, J. Rodriguez, and M. Perez, "A Survey on Cascaded Multilevel Inverters," *IEEE Transactions on Industrial Electronics*, vol. 57, no. 7, pp. 2197–2206, 2010.
- [8] A. Sannino, M. Bollen, and J. Svensson, "Power-Electronic Solutions to Power Quality Problems," *Electric Power Systems Research*, vol. 66, pp. 71–82, 2003.
- [9] K. Vijaya Kumar, M. R. Kumar, and P. R. Rao, "Modeling and Simulation of Electric Vehicle Charging Station Controlled by Fuzzy Logic Controller," *International Journal of Electrical Power and Energy Systems*, vol. 85, pp. 118–126, 2017.
- [10] M. Ehsani, Y. Gao, and S. E. Gay, *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles*, CRC Press, 2005.
- [11] A. Emadi, K. Rajashekara, S. Williamson, and S. Lukic, "Topological Overview of Hybrid Electric and Fuel Cell Vehicular Power System Architectures," *IEEE Transactions on Vehicular Technology*, vol. 54, no. 3, pp. 763–770, 2005.
- [12] G. Masters, *Renewable and Efficient Electric Power Systems*, Wiley, 2013.
- [13] M. A. Quddus, M. A. Hannan, and A. Mohamed, "Electric Vehicle Charging Station

Expansion with Renewable Energy Integration and Vehicle-to-Grid Technology,” *Energy*, vol. 120, pp. 256–267, 2017.

[14] R. W. Erickson and D. Maksimovic, *Fundamentals of Power Electronics*, Springer, 2001.

[15] T. Markel et al., “Communication and Control of Electric Drive Vehicles Supporting Renewables,” *IEEE Power and Energy Society*, 2010.

[16] H. Rashid, *Power Electronics: Circuits, Devices, and Applications*, Pearson, 2014.

[17] S. Chowdhury and S. P. Chowdhury, *Microgrids and Active Distribution Networks*, IET Press, 2009.

[18] N. Mohan, T. Undeland, and W. Robbins, *Power Electronics: Converters, Applications and Design*, Wiley, 2003.

[19] J. Momoh, *Smart Grid: Fundamentals of Design and Analysis*, Wiley, 2012.

[20] S. Choudhury, P. Das, and S. Saha, “Advancements in Multilevel Inverter Technology for Renewable Energy Applications,” *Renewable Energy Journal*, vol. 145, pp. 128–140, 2019.

[21] A. Khaligh and O. Onar, *Energy Harvesting: Solar, Wind, and Ocean Energy Conversion Systems*, CRC Press, 2017.

[22] J. Larminie and J. Lowry, *Electric Vehicle Technology Explained*, Wiley, 2012.

[23] S. Tie and C. Tan, “A Review of Energy Sources and Energy Management System in Electric Vehicles,” *Renewable and Sustainable Energy Reviews*, vol. 20, pp. 82–102, 2013.

[24] P. T. Krein, *Elements of Power Electronics*, Oxford University Press, 2015.

[25] B. Singh, B. N. Singh, A. Chandra, K. Al-Haddad, A. Pandey, and D. Kothari, “A Review of Single-Phase Improved Power Quality AC–DC Converters,” *IEEE Transactions on Industrial Electronics*, vol. 50, no. 5, pp. 962–981, 2003.

[26] M. Shahidehpour and Y. Wang, *Communication and Control in Electric Power Systems*, Wiley, 2003.

[27] L. Zhang, T. Brown, and S. Samuelsen, “Evaluation of Charging Infrastructure Requirements and Operating Costs for Plug-in Electric Vehicles,” *Journal of Power Sources*, vol. 240, pp. 515–524, 2013.

[28] Y. He, B. Venkatesh, and L. Guan, “Optimal Scheduling for Charging and Discharging of Electric Vehicles,” *IEEE Transactions on Smart Grid*, vol. 3, no. 3, pp. 1095–1105, 2012.

[29] S. Shao, M. Pipattanasomporn, and S. Rahman, “Demand Response as a Load Shaping Tool in an Intelligent Grid,” *IEEE Transactions on Smart Grid*, vol. 2, no. 4, pp. 624–631, 2011.

- [30] J. Guerrero, P. Loh, T. Lee, and M. Chandorkar, "Advanced Control Architectures for Intelligent Microgrids," *IEEE Transactions on Industrial Electronics*, vol. 60, no. 4, pp. 1254–1262, 2013.
- [31] C. C. Chan, "The State of the Art of Electric and Hybrid Vehicles," *Proceedings of the IEEE*, vol. 90, no. 2, pp. 247–275, 2002.
- [32] J. Hu, S. You, and M. Lind, "Charging Coordination of Electric Vehicles for Grid Support," *IEEE Transactions on Smart Grid*, vol. 8, no. 2, pp. 1001–1010, 2017.
- [33] R. C. Green, L. Wang, and M. Alam, "The Impact of Plug-in Hybrid Electric Vehicles on Distribution Networks," *IEEE Transactions on Smart Grid*, vol. 1, no. 3, pp. 247–254, 2010.
- [34] S. Deilami, A. Masoum, P. Moses, and M. Masoum, "Real-Time Coordination of Plug-in Electric Vehicle Charging in Smart Grids," *IEEE Transactions on Smart Grid*, vol. 2, no. 3, pp. 456–467, 2011.
- [35] M. Yilmaz and P. T. Krein, "Review of Charging Power Levels and Infrastructure for Plug-in Electric and Hybrid Vehicles," *IEEE Transactions on Power Electronics*, vol. 28, no. 5, pp. 2151–2169, 2013.
- [36] K. Clement-Nyns, E. Haesen, and J. Driesen, "The Impact of Charging Plug-in Hybrid Electric Vehicles on a Residential Distribution Grid," *IEEE Transactions on Power Systems*, vol. 25, no. 1, pp. 371–380, 2010.
- [37] T. Markel et al., "Communication and Control of Electric Drive Vehicles Supporting Renewables," *IEEE Power and Energy Society General Meeting*, pp. 1–8, 2010.
- [38] S. Rahman and G. Shrestha, "An Investigation into the Impact of Electric Vehicle Load on the Electric Utility Distribution System," *IEEE Transactions on Power Delivery*, vol. 8, no. 2, pp. 591–597, 1993.
- [39] J. Tomic and W. Kempton, "Using Fleets of Electric Drive Vehicles for Grid Support," *Journal of Power Sources*, vol. 168, pp. 459–468, 2007.
- [40] M. Etezadi-Amoli, K. Choma, and J. Stefani, "Rapid-Charge Electric Vehicle Stations," *IEEE Transactions on Power Delivery*, vol. 25, no. 3, pp. 1883–1887, 2010.
- [41] A. Khaligh and Z. Li, "Battery, Ultracapacitor, Fuel Cell, and Hybrid Energy Storage Systems for Electric Vehicles," *IEEE Transactions on Vehicular Technology*, vol. 59, no. 6, pp. 2806–2814, 2010.
- [42] B. Bilgin, A. Emadi, and M. Krishnamurthy, "Comprehensive Evaluation of Electric Motor Drives for EV Applications," *IEEE Transactions on Vehicular Technology*, vol. 62, no. 8, pp. 3267–3279, 2013.
- [43] M. Kazerani and P. Venkatesh, "Renewable Energy Integration in Electric Vehicle Charging Stations," *IEEE Transactions on Sustainable Energy*, vol. 4, no. 3, pp. 684–692,

2013.

[44] H. Turker, S. Bacha, D. Chatroux, and A. Hably, “Low-Voltage Transformer Loss of Life Evaluating Electric Vehicle Charging Impact,” *IEEE Transactions on Power Delivery*,

vol. 27, no. 3, pp. 1774–1782, 2012.

[45] W. Kempton and J. Tomic, “Vehicle-to-Grid Power Implementation: From Stabilizing the Grid to Supporting Large-Scale Renewable Energy,” *Journal of Power Sources*, vol. 144, pp. 280–294, 2005.

[46] N. Liu, J. Zhang, and L. Wang, “Energy Management for EV Charging Stations with Renewable Energy Integration,” *IEEE Transactions on Smart Grid*, vol. 9, no. 2, pp. 1210–1218, 2018.

[47] Y. He, B. Venkatesh, and L. Guan, “Optimal Scheduling for Charging and Discharging of Electric Vehicles,” *IEEE Transactions on Smart Grid*, vol. 3, no. 3, pp. 1095–1105, 2012.

[48] S. Shao, M. Pipattanasomporn, and S. Rahman, “Demand Response as a Load Shaping Tool in an Intelligent Grid,” *IEEE Transactions on Smart Grid*, vol. 2, no. 4, pp. 624–631, 2011.

[49] A. Dubey and S. Santoso, “Electric Vehicle Charging on Residential Distribution Systems: Impacts and Mitigations,” *IEEE Access*, vol. 3, pp. 1871–1893, 2015.

[50] J. Guerrero, P. Loh, T. Lee, and M. Chandorkar, “Advanced Control Architectures for Intelligent Microgrids,” *IEEE Transactions on Industrial Electronics*, vol. 60, no. 4, pp. 1254–1262, 2013.

APPENDIX

SOURCE CODE FOR SMART EV CHARGING SYSTEM

This appendix presents the software programs developed for the implementation of the smart charging station system. The system consists of three major software components: the ESP32 microcontroller program for voltage monitoring and data transmission, the web-based dashboard interface for monitoring charging parameters, and the Node.js server program used for handling communication between the hardware system and the dashboard. These programs enable real-time monitoring of solar panel voltage, DC charger voltage, and charging status through a web-based dashboard.

A.1 ESP32 Program for Voltage Monitoring and Data Transmission

The following program was developed using the ESP32 microcontroller platform. The ESP32 reads the analog voltage values from the solar panel and DC charger using ADC pins. The measured voltage values are converted into actual voltage levels and transmitted to a server through WiFi using HTTP POST requests. The program continuously updates the voltage values every few seconds to provide real-time monitoring of the charging system.

```
#include <WiFi.h>
#include <HTTPClient.h>
// WiFi credentials
const char* ssid = "Redmi 12";
const char* password = "puji1234";
// Server endpoint
const char* server = "http://192.168.187.76:3000/api/voltage";
// Voltage sensor analog pins
#define SOLAR_PIN 34
#define DC_PIN 35
#define REF_VOLTAGE 3.3
```

```

#define ADC_RESOLUTION 4095.0

#define VOLTAGE_SCALE 5.0

void setup() {
  Serial.begin(115200);
  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
    Serial.print(".");
  }
  Serial.println("\nWiFi connected.");
}

void loop() {
  int raw_solar = analogRead(SOLAR_PIN);
  int raw_dc = analogRead(DC_PIN);

  float solar_voltage = (raw_solar * REF_VOLTAGE / ADC_RESOLUTION) *
    VOLTAGE_SCALE;

  float dc_voltage = (raw_dc * REF_VOLTAGE / ADC_RESOLUTION) *
    VOLTAGE_SCALE;

  Serial.printf("Solar Voltage: %.2f V, DC Voltage: %.2f V\n", solar_voltage,
    dc_voltage);

  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;
    http.begin(server);
    http.addHeader("Content-Type", "application/json");
    String jsonPayload = "{\"solar\": " + String(solar_voltage,2) +
      ", \"dc\": " + String(dc_voltage,2) + "}";
    int httpResponseCode = http.POST(jsonPayload);
    http.end();
  }
}

```

```
delay(3000);  
}
```

A.2 Smart Charging Station Dashboard (Web Interface)

The dashboard interface is developed using HTML, CSS, and JavaScript. It provides a real-time visualization of the charging system parameters including solar panel voltage, DC charger voltage, charging source status, and battery status. The interface periodically retrieves updated data from the server and displays the charging status using visual indicators.

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
  
<title>Smart Charging Station Dashboard</title>  
  
</head>  
  
<body>  
  
<h1>Smart Charging Station Dashboard</h1>  
  
<div id="solar">Solar Panel Voltage</div>  
  
<div id="dc">DC Charger Voltage</div>  
  
<div id="charging-source"></div>  
  
<div id="battery-status"></div>  
  
<script>  
async function fetchData() {  
  const response = await fetch('/data.json');  
  const data = await response.json();  
  document.getElementById('solar').innerHTML =  
  "Solar Panel Voltage: " + data.solar + " V";  
  document.getElementById('dc').innerHTML =  
  "DC Charger Voltage: " + data.dc + " V";  
}  
setInterval(fetchData,10000);
```


</script>

</body>

</html>

A.3 Node.js Server Program

The server program was developed using Node.js and Express framework. It receives voltage data from the ESP32 microcontroller through HTTP requests and stores the data in a JSON file. The stored data is then accessed by the web dashboard for real-time monitoring of the charging system.

```
const express = require('express');
const fs = require('fs');
const path = require('path');
const app = express();
const port = 3000;
const dataFile = path.join(__dirname, 'public', 'data.json');
app.use(express.static(path.join(__dirname, 'public')));
app.use(express.json());
app.get('/data.json',(req,res)=>{
  const data = fs.readFileSync(dataFile);
  res.setHeader('Content-Type','application/json');
  res.send(data);
});
app.post('/api/voltage',(req,res)=>{
  const {solar,dc} = req.body;
  if(typeof solar === 'number' && typeof dc === 'number'){
    const data = {solar,dc,timestamp:new Date()};
    fs.writeFileSync(dataFile,JSON.stringify(data));
    res.json({success:true});
  }
```

```
else{  
  res.status(400).json({error:"Invalid data"});  
  
}  
});  
app.listen(port,()=>{  
  console.log(`Server running at http://localhost:${port}`);  
});
```

PUBLICATION DETAILS

[1] M. Siva Rama Ganesh, G. Sailaja Ramya Krishna, V. Sri Pujitha, B. Chandini, and M. Monanjali, “AI-Controlled Multi-Source Power Integration in Sustainable EV Charging Stations,” presented at the 2026 International Conference on Electric Power and Renewable Energy (EPREC-2026) organized by the Department of Electrical Engineering published in scopus indexed conference, Indian Institute of Technology Bhilai, held during 2–4 January 2026, India.

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